Experiment 202: Data Model

Focused comparison of full Muon and hard-top-k Adaptive Muon on WV+Wout. K varies while every sample retains exactly five distinct signal features.

1. Signal and nuisance dictionaries

$$d = K_1 + K_2, \qquad \mathcal{O} = [\, M \;\; M_\perp \,]$$

$$M^\top M = I, \quad M_\perp^\top M_\perp = I, \quad M^\top M_\perp = 0$$

A Gaussian d1-by-d1 matrix is orthogonalized by QR. The first K1 columns of M are M1 features, the next K2 are M2 features, and Mperp spans the orthogonal nuisance subspace.

2. Fixed five-signal construction

$$X = [x_1, \ldots, x_K]^\top \in \mathbb{R}^{K \times d_1}$$

- Every sample contains exactly 5 signal tokens.
- Each signal slot is M1 with probability 0.8 and M2 with probability 0.2.

$$n_{\mathrm{M1}} \sim \mathrm{Binomial}(5, 0.8), \quad n_{\mathrm{M2}} = 5 - n_{\mathrm{M1}}$$

$$\mathbb{E}[n_{\mathrm{M1}}] = 4, \qquad \mathbb{E}[n_{\mathrm{M2}}] = 1$$

- Features are sampled without replacement inside each group, so all five signal features are distinct.
- The remaining K-5 tokens are independently sampled unit-norm columns of Mperp.
- Signal and Mperp amplitudes are both 1; there is no additive Gaussian noise.

3. K sweep

K = 18, 20, 23, 25, 28, 30. The corresponding Mperp counts are 13, 15, 18, 20, 23, 25. K changes only nuisance-token count; signal count remains five.

Experiment 202: Label and Network

1. Presence label and binary teacher

$$z \in \{0,1\}^d, \qquad |z|_0 = 5$$

z marks which signal features occur in the sequence. Mperp tokens activate no label coordinate.

$$W_{\text{true}} \in \{0,1\}^{C \times d}, \qquad y = W_{\text{true}} z$$

Each teacher column has exactly round(0.1 C) ones. The target is the sum of the five active binary code columns. There is no M1/M2 teacher reweighting.

2. Softmax self-attention

$$Q = XW_Q, \qquad K_{\text{att}} = XW_K$$

$$A = \text{softmax}\left(QK_{\text{att}}^{\top}/\sqrt{d_{qk}}\right)$$

$$\widetilde{X} = AX$$

3. Value map and readout

$$V = \text{ReLU}(\widetilde{X}W_V)$$

$$h = \sum_{k=1}^{K} V_k, \qquad \widehat{y} = W_{\text{out}} h$$

There is no additional MLP. Pooling is a sum, not a mean, so the activation/readout scale may depend on K.

4. Exact shapes

$$W_Q, W_K, W_V \in \mathbb{R}^{100 \times 100}$$

$$W_{\text{out}} \in \mathbb{R}^{50 \times 100}$$

$$d_1 = 100, \quad d = 20, \quad K_1 = K_2 = 10, \quad C = 50$$

Experiment 202: Loss, Optimizer, and Evaluation

1. L2 regression loss

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{2} |\widehat{y}^{(i)} - y^{(i)}|_2^2$$

Training uses N=32, 1000 steps, learning rate 0.03, and seed 0. Normalized loss is post-processing only.

2. Full Muon

$$B_t = 0.95 B_{t-1} + G_t, \qquad B_t = U \operatorname{diag}(\sigma) V^\top$$

$$\Delta W_t = U V^\top$$

3. Hard-top-k Adaptive Muon

WQ and WK remain full Muon. WV and Wout estimate rank from the momentum-buffer spectrum. The threshold is tau=5 times the median of the bottom half of the singular values.

$$k_{\text{raw}} = \#\{i : \sigma_i > 5\sigma_{\text{noise}}\}$$

$$\bar{k}_t = 0.005 k_{\text{raw},t} + 0.995 \bar{k}_{t-1}$$

$$\Delta W_t = U_{:,1:k_t} V^\top_{:,1:k_t}$$

There is no warmup. All singular directions after k receive zero update.

4. Matched-K evaluation

- Source: five random M1/M2 features, expected ratio 4:1.
- Only M1: five distinct M1 features.
- Only M2: five distinct M2 features.
- Mixed: exactly four M1 features and one M2 feature.

Every evaluation set contains the same K-5 Mperp count as training and has 5000 fixed held-out samples. The reported late statistic is mean loss over steps 800-1000.